Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/14/2007
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Pool Boiling Heat Transfer Mechanisms in Microgravity and Partial Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/22/2003	**End Date:**	10/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2007 per PI (11/06). Note: extended 4 more years		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA26A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of the proposed research is to determine the effects of microgravity on nucleate boiling heat transfer. An understanding of microgravity effects on boiling mechanisms is critical to the proper design of phase change heat removal equipment for use in space-based applications. Experiments to date have shown that stable, subcooled boiling on flat plates in microgravity environments is possible. It is presently thought that coalescence and condensation of bubbles away from the heater cause oscillatory motions within the surrounding liquid, thereby providing a continual supply of liquid from the bulk of the pool to the base of the bubble just above the heater. Almost all research pertaining to boiling in microgravity environments, however, has thus far been either of a qualitative nature (photographic studies) with some average wall heat flux/wall temperature measurements, analytical work, or numerical simulations. In the studies where the heat transfer coefficients were measured, the heated surface was always comparable to or much larger than the bubble sizes, so only average heat transfer rates over the entire heated surface were obtained. Very little experimental data is available regarding the local heat transfer rates under and around the bubbles as they grow and depart from the surface. Such information, especially when correlated with visual studies, can provide much needed information regarding the relevant wall heat transfer mechanisms during the bubble departure cycle by pinpointing when and where in the cycle large amounts of heat are removed. Microgravity effects on critical heat flux levels (the maximum heat flux level that can be attained without a catastrophic increase in heater temperature) is another very important area that must be addressed if boiling is to be used reliably as a heat removal mechanism. Although critical heat flux is known to decrease with decreasing gravity, very little quantitative information is available due to the use of constant wall heat flux heaters. A four-year project with two primary objectives is proposed: 1) to determine the relative contributions of various heat transfer mechanisms to the overall heat flux in subcooled, nucleate pool boiling on a flat plate in microgravity environments, and 2) to obtain quantitative data on the effects of microgravity on critical heat flux levels. The research will results in better understanding of the mechanisms of boiling heat transfer, and should enable the development of more efficient two-phase heat exchangers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research has shown that transient conduction is the dominant heat transfer mechanism in boiling of refrigerants-like fluids.
Task Progress:	The focus of the research the past year was to obtaining single bubble data using a fluid whose properties were quite different from FC-72 to test whether or not the transient conduction heat transfer mechanisms we found with FC-72 can be applied to other fluids. Cooper and Lloyd (1969) found that microlayer evaporation was the primary heat transfer mechanism using toluene and isopropyl alcohol–these fluids require much larger amounts of energy to vaporize the liquid compared to the heat required for sensible heating than for FC-72 which has a relatively low heat of vaporization. Initial tests with ethanol and water revealed that these fluids cause the microheater array to corrode and could not be used. Pentane was decided on as the working fluid after additional tests found no degradation of the array or the wire connecting the leads to the carrier. The circuits required to control the heater array using pentane were designed, constructed, and debugged. Heat transfer coefficient data using pentane along with visual images have been obtained, and data has been reduced. Because the data did not include bubble nucleating towards the center of the array, additional data is being acquired. Numerical and analytical studies on the origin of thermocapillary convection during subcooled boiling was investigated. A qualitative study was performed on the effects of dissolved gas content, bubble shape and size, and heat transfer coefficient on the strength of thermocapillary convection. Due to the presence of different complex and interrelated mechanisms, the individual mechanisms were decoupled and their overall effect on thermocapillary convection was studied. The results indicate that variations in heat transfer coefficient and the shape and the size of the bubble play a major role in the development of thermocapillary motion. The experiments on electric field effects on boiling heat transfer in cooperation with Cila Herman at Johns Hopkins University were continued. Data was obtained with the microheater array facing downward so that the bubble was pushed against the heater array. Boiling curves were obtained for 2x2 and 3x3 arrays of heaters. The electric field was observed to be very effective in removing the primary bubble from the heated surface, and allowing satellite bubble nucleation, growth, and departure to occur. The heat transfer with electric field was much higher than without electric field.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Articles in Peer-reviewed Journals	Sathyamurthi V, Banerjee D, Sakamoto H, Kim J. "Measurement of the fractal order of wall void fraction during nucleate boiling." International Journal of Heat and Fluid Flow, 2008 Feb;29(1):207-18. http://dx.doi.org/10.1016/j.ijheatfluidflow.2007.03.009 , Feb-2008
Articles in Peer-reviewed Journals	Liu Z, Herman C, Kim J. "Heat transfer and bubble detachment in subcooled pool boiling from a downward-facing microheater array in a nonuniform electric field." Annals of the New York Academy of Sciences. 2009 Apr;1161:182-91. PubMed PMID: 19426316 ; http://dx.doi.org/10.1111/j.1749-6632.2008.04331.x , Apr-2009
Articles in Peer-reviewed Journals	Raj R, Kim J. "Thermocapillary convection during subcooled boiling in reduced gravity environments." Annals of the New York Academy of Sciences. 2009 Apr;1161:173-81. PubMed PMID: 19426315 ; http://dx.doi.org/10.1111/j.1749-6632.2008.04327.x , Apr-2009
Papers from Meeting Proceedings	Liu Z, Herman C, Kim J. "Heat transfer and bubble detachment in subcooled pool boiling from a microheater array under the effect of a nonuniform electric field." ASME/JSME/Interpak Conference, Vancouver, CA, July, 2007. Proceedings of the ASME/JSME/Interpak Conference, July, 2007. , Jul-2007